

SymPy Bug Card

Bug 16: incorrect Laplace transform of Heaviside($a - t$)

Task. Compute the Laplace transform of a closing step with real cutoff a .

$$\mathcal{L}\{\text{Heaviside}(a - t)\}(s), \quad a \in \mathbb{R}, \ s > 0.$$

SymPy's answer for symbolic a .

$$\mathcal{L}_{\text{SymPy}}\{\text{Heaviside}(a - t)\}(s) = \frac{e^{-as}}{s}.$$

Correct answer.

$$\mathcal{L}\{\text{Heaviside}(a - t)\}(s) = \begin{cases} \frac{1 - e^{-as}}{s}, & a > 0, \\ 0, & a \leq 0. \end{cases}$$

Explanation. The Laplace integral runs over $[0, \infty)$: this step contributes on $[0, a)$ when $a > 0$, and on no interval when $a \leq 0$. That gives the piecewise result above, not e^{-as}/s .

Likely root cause. In `sympy/integrals/laplace.py`, the symbolic path returns e^{-as}/s guarded by the impossible condition $(a > 0) \wedge (a < 0) \wedge \text{Ne}(1/a, 0)$; `noconds=True` discards that guard.

Artifact (GitHub).

[bug-report/bug-016-incorrect-laplace-transform-of-heaviside-a-t/](#)